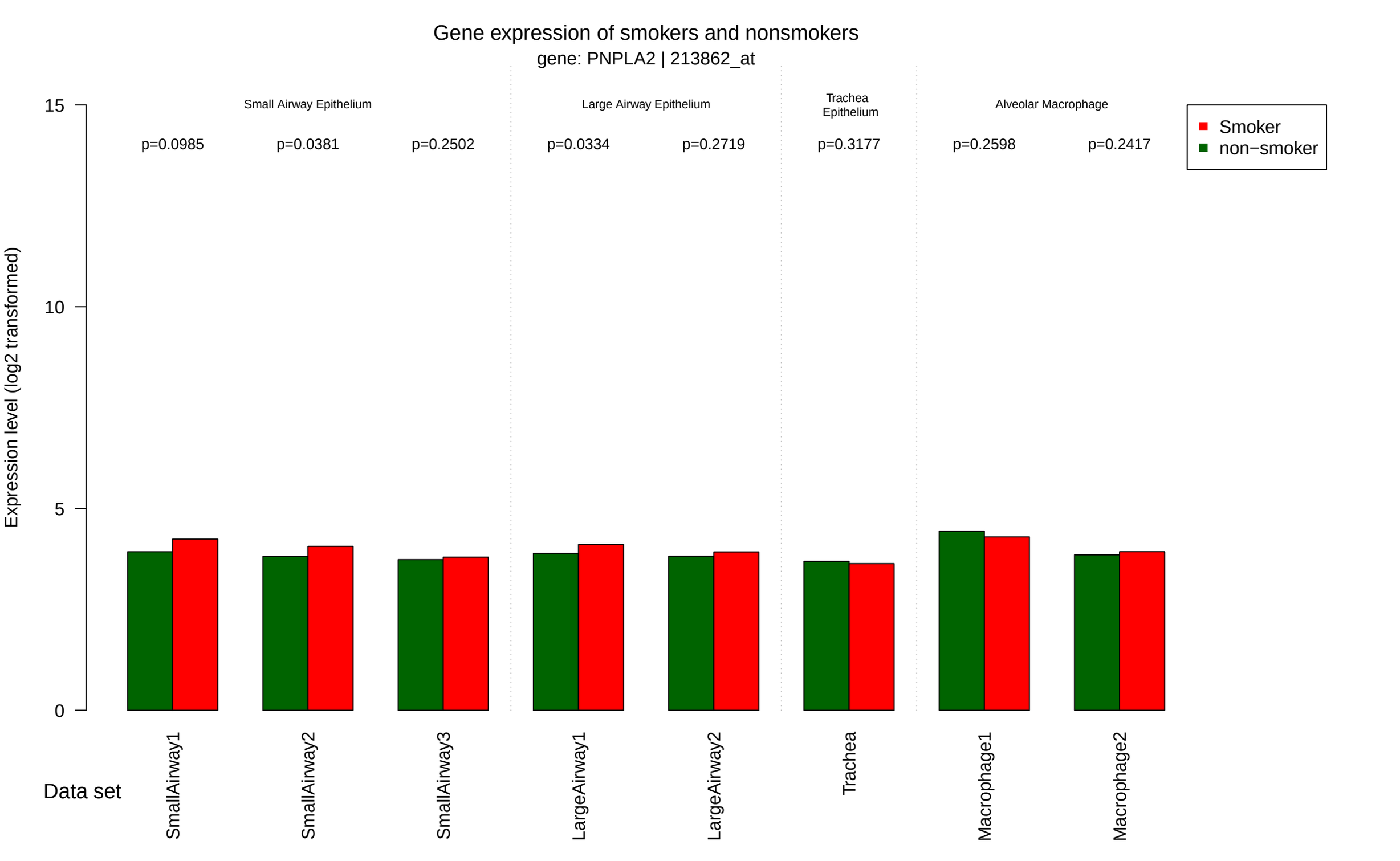
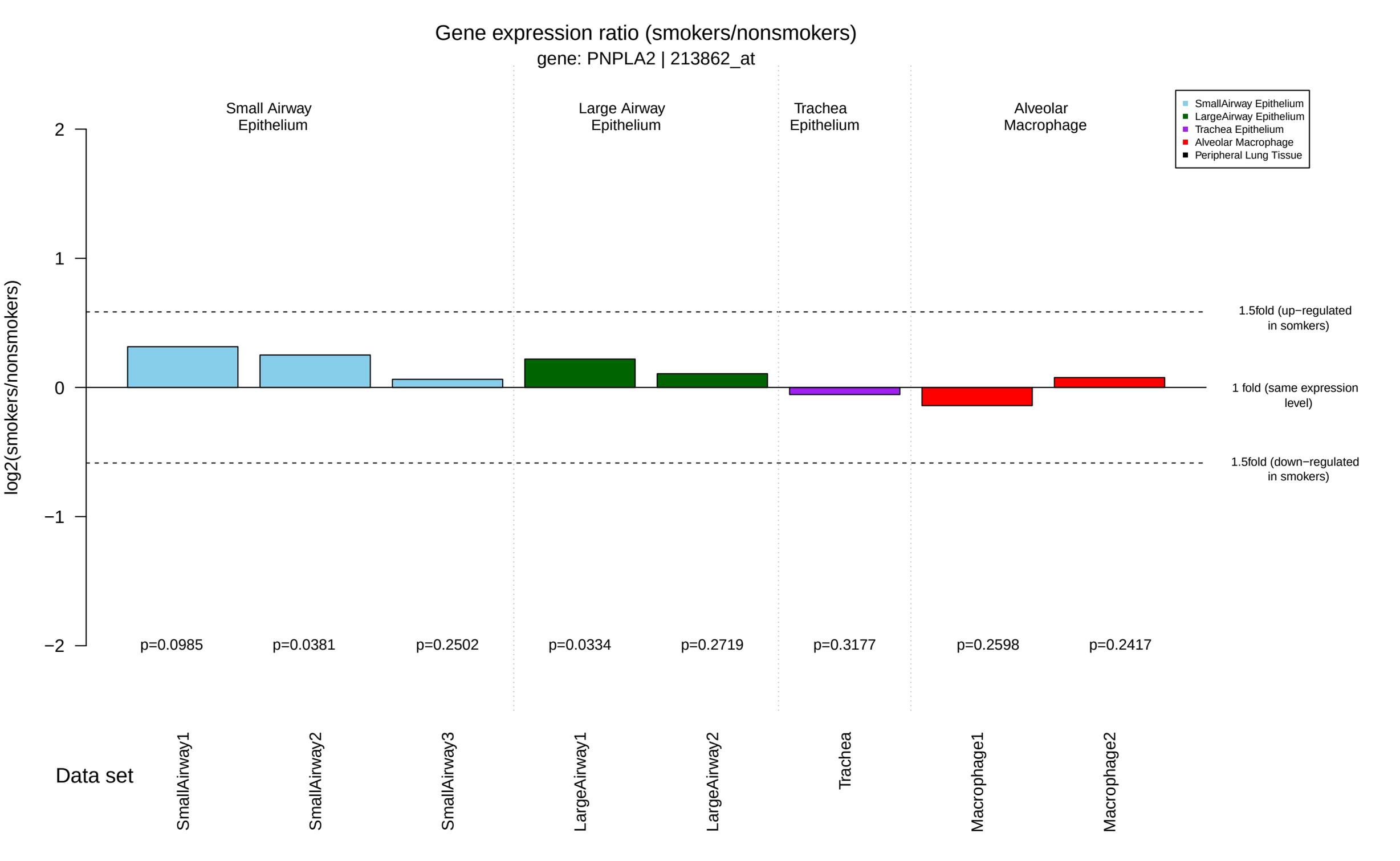
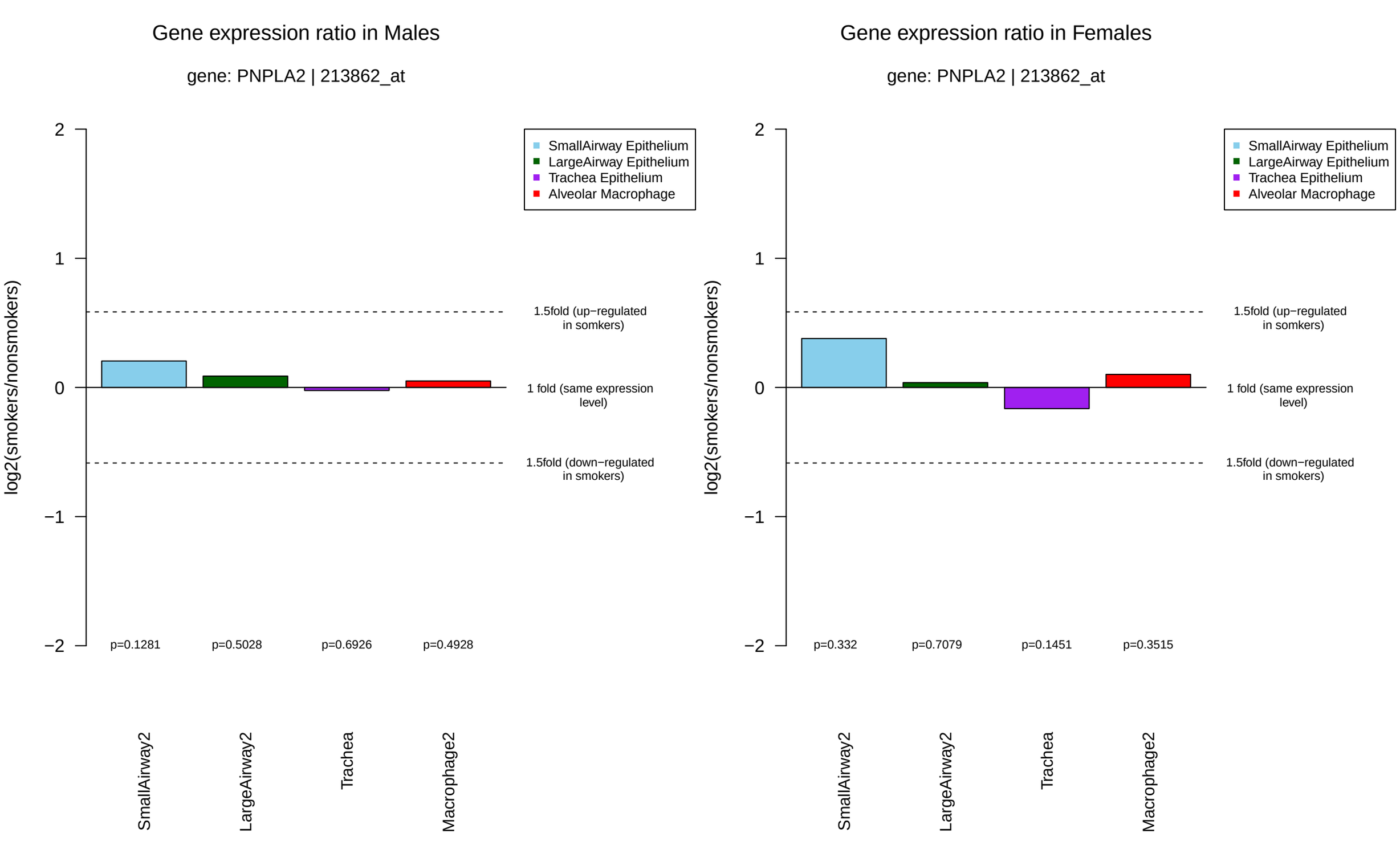
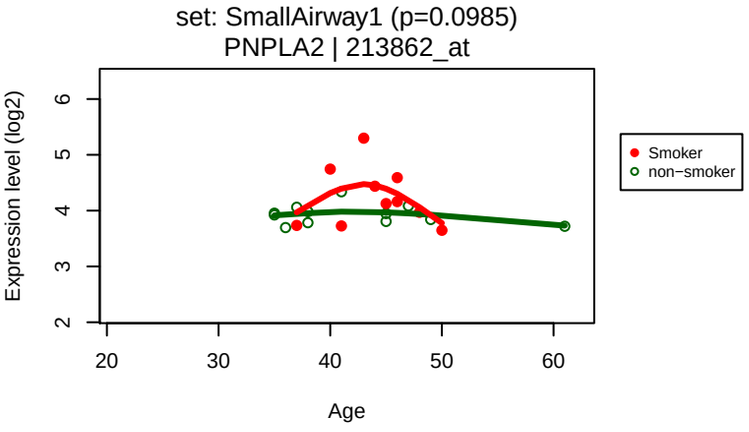
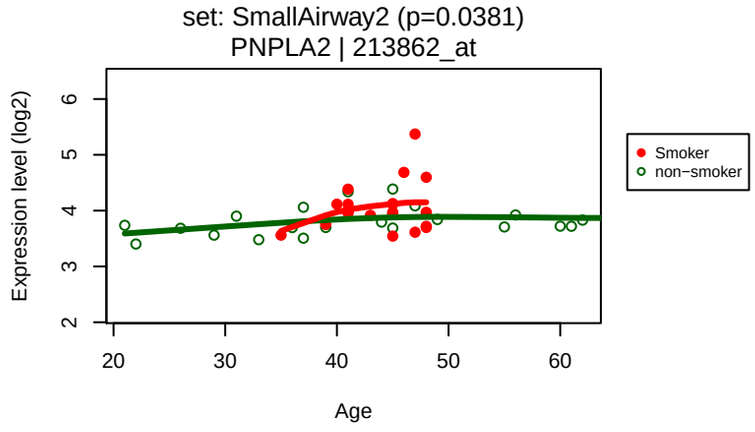
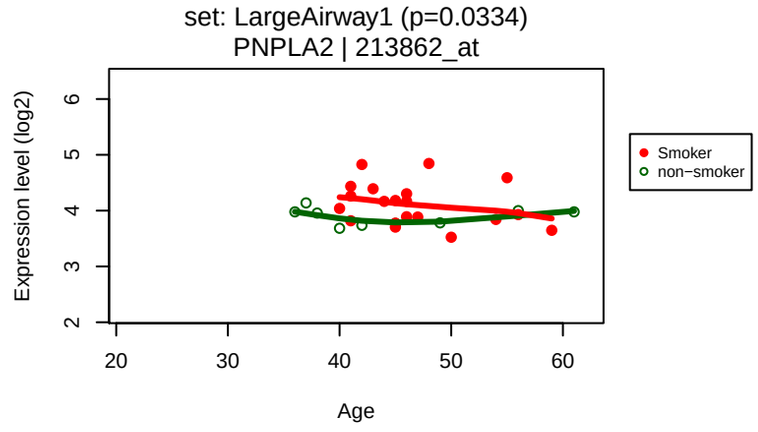

Gene expression of smokers and nonsmokers
gene: PNPLA2 | 213862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0985
p=0.0381
p=0.2502
p=0.0334
p=0.2719
p=0.3177
p=0.2598
p=0.2417
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

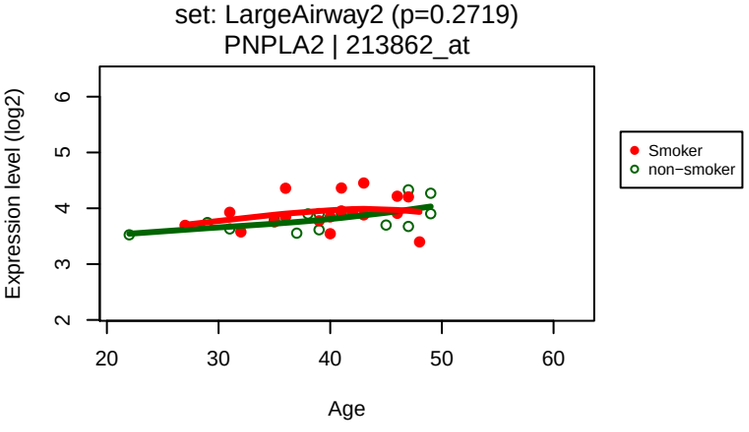

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA2 | 213862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0985
p=0.0381
p=0.2502
p=0.0334
p=0.2719
p=0.3177
p=0.2598
p=0.2417
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

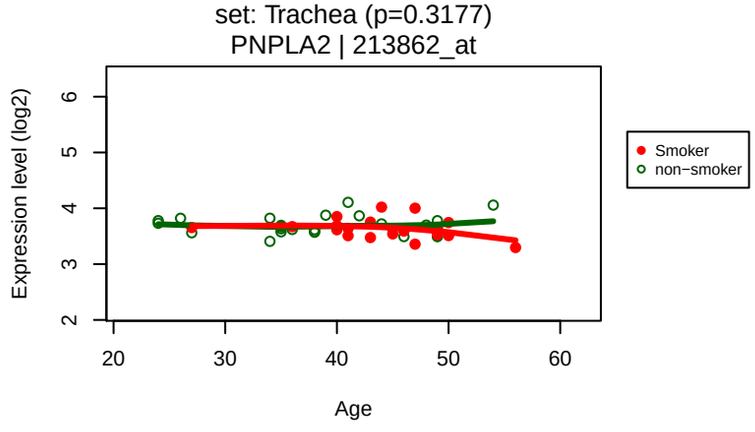

Gene expression ratio in Males
gene: PNPLA2 | 213862_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1281
p=0.5028
p=0.6926
p=0.4928
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA2 | 213862_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.332
p=0.7079
p=0.1451
p=0.3515
SmallAirway2
LargeAirway2
Trachea
Macrophage2

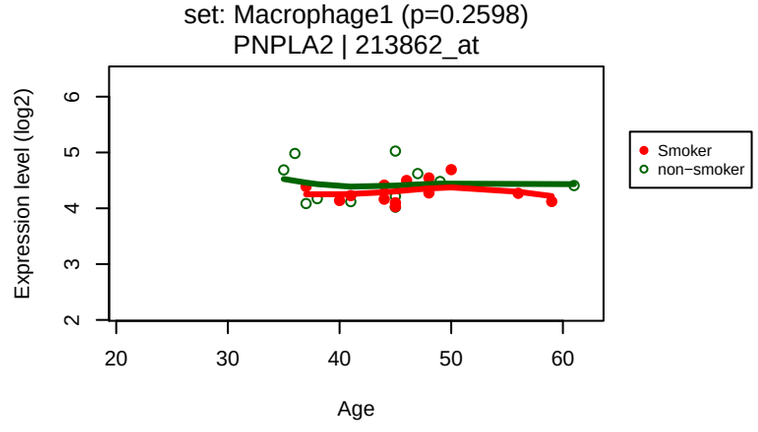

set: SmallAirway1 (p=0.0985)
PNPLA2 | 213862_at
Expression level (log2)
Age
Smoker
non−smoker

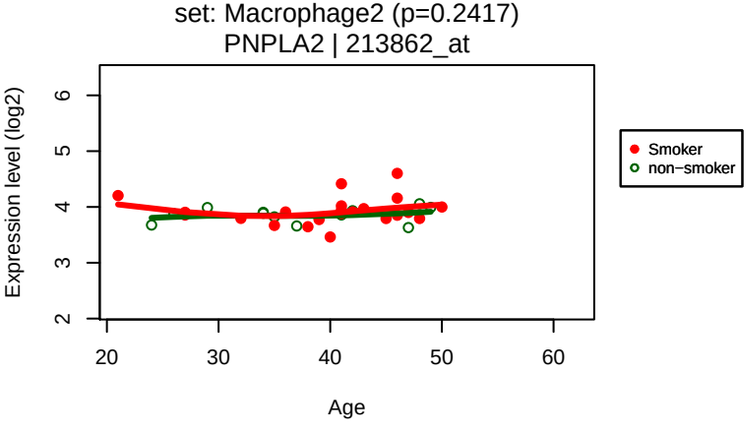

set: SmallAirway2 (p=0.0381)
PNPLA2 | 213862_at
Expression level (log2)
Age
Smoker
non−smoker

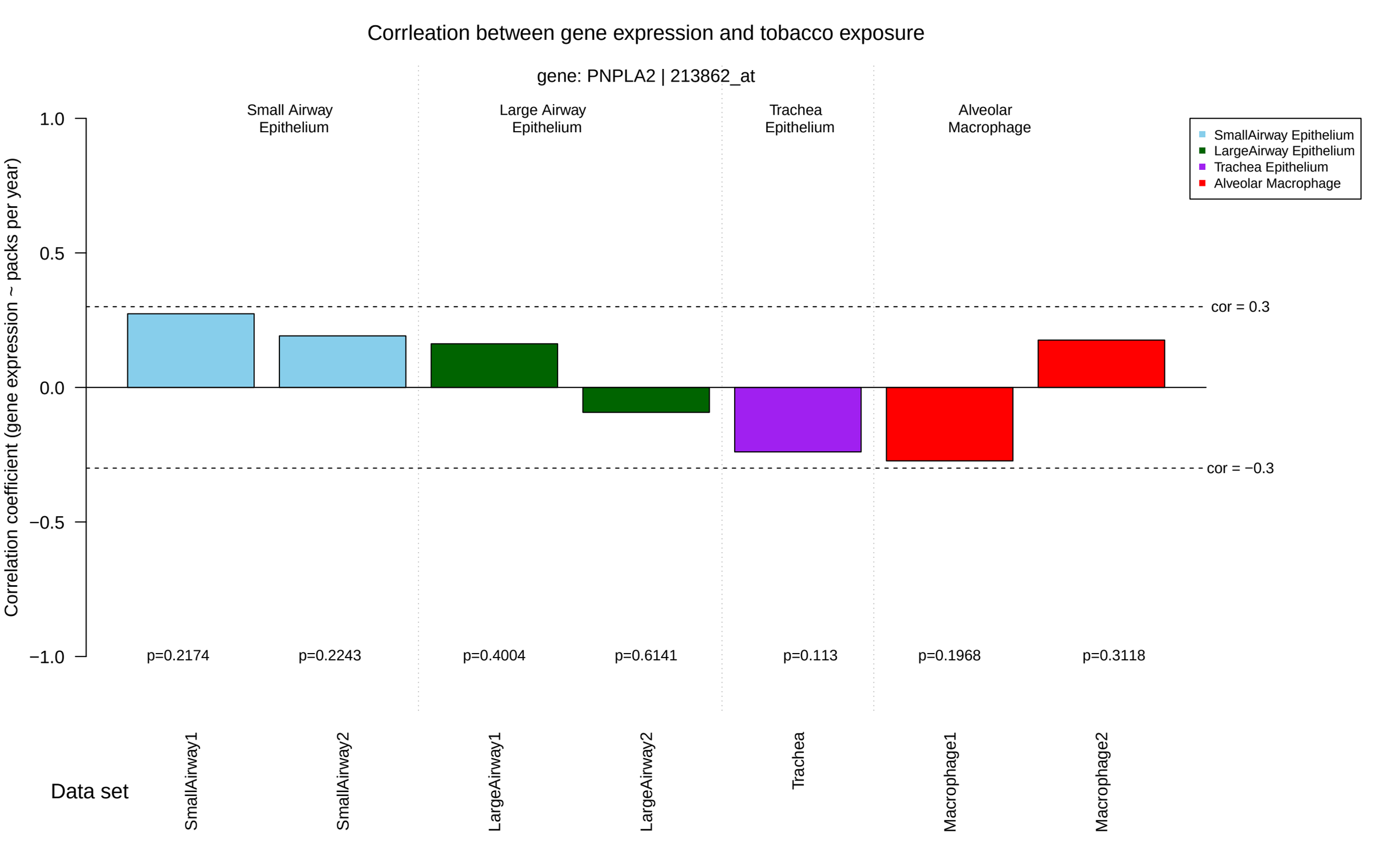

set: LargeAirway1 (p=0.0334)
PNPLA2 | 213862_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2719)
PNPLA2 | 213862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3177)
PNPLA2 | 213862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2598)
PNPLA2 | 213862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2417)
PNPLA2 | 213862_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA2 | 213862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2174
p=0.2243
p=0.4004
p=0.6141
p=0.113
p=0.1968
p=0.3118
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2